

Advanced Clustering Analysis of NHANES 2017–2018 Health Data

*Comparative Evaluation of K-Means, DBSCAN, and Gaussian Mixture
Models*

Final Report

DATA 542 – Advanced Clustering Methods

Spring 2026

Prepared by:

Banira Acharya Dhakal
Segan Foster
Monique Handsom

Submitted to:

Professor John Koomson
Department of Data Science
Central Connecticut State University

Date Submitted: May 2026

1. Introduction

Clustering is an unsupervised machine learning technique used to identify natural groupings in data without predefined labels. In healthcare, it can help reveal hidden patterns and risk profiles among individuals. This project applies clustering techniques to a subset of the National Health and Nutrition Examination Survey (NHANES) dataset. The selected variables capture key aspects of individuals' demographic, physiological, and metabolic characteristics, including age, income level, body mass index (BMI), blood pressure, and glycohemoglobin (HbA1c). The goal is to identify meaningful health-related clusters, interpret their characteristics, and compare different clustering approaches.

Clustering fits this health data problem because the dataset does not have predefined health-risk groups. Individuals vary across age, income, BMI, blood pressure, and HbA1c, and clustering helps identify natural patterns across these variables. This makes it useful for finding meaningful health profiles, such as diabetes-related risk, cardiovascular risk, obesity-related risk, and lower-risk groups.

2. Exploratory Data Analysis

Exploratory data analysis was conducted to understand the structure and quality of the selected NHANES variables before applying clustering methods. First, summary statistics were generated to examine the central tendency and spread of the variables.

Missing values were then identified by calculating the number and percentage of missing observations for each feature, and a bar chart was used to visualize missingness across the dataset. To address missing data, median imputation was applied to numeric variables because the median is less affected by outliers than the mean.

Detailed descriptive statistics, including mean, median, variance, and skewness, were calculated. Histograms were used to examine variable distributions, and boxplots were used to identify potential outliers.

A correlation matrix was also created to examine relationships among variables. The analysis showed a high correlation between BMI and waist circumference, so waist was removed to avoid redundancy. The sex variable was also excluded because it is categorical and not suitable for direct use in distance-based clustering.

The remaining variables were standardized before clustering so that all features contributed equally to the analysis.

2.1 Detailed Summary Statistics

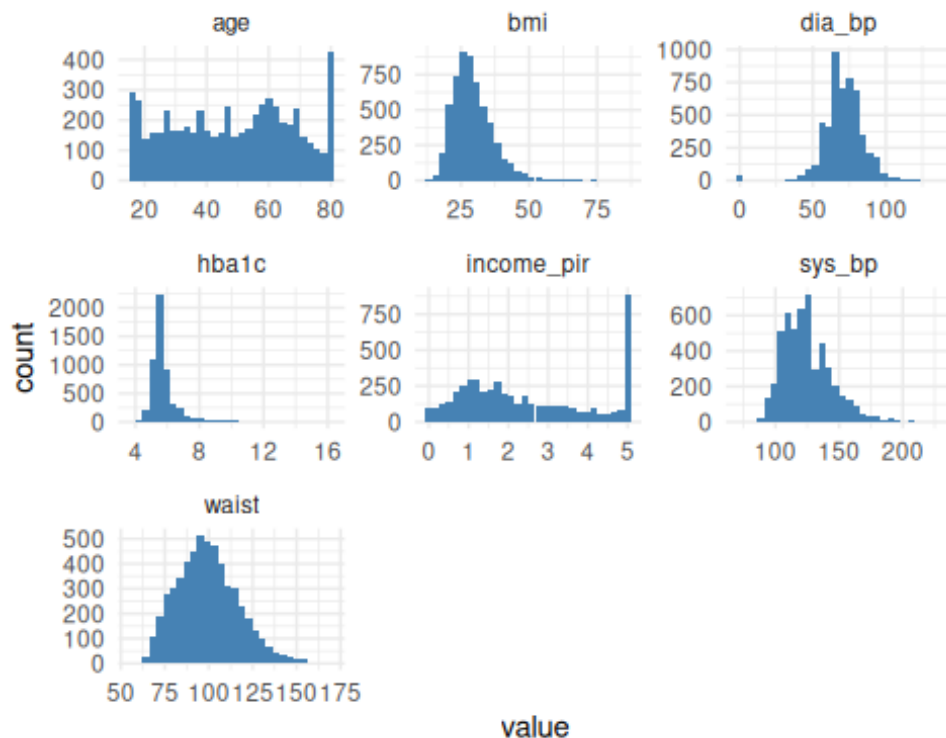
Summary Statistics Table

Variable	mean	median	var	skew
age	48.130	49.00	383.592	-0.040
income_pir	2.512	2.07	2.584	0.353

bmi	29.462	28.30	55.752	1.245
waist	99.336	98.10	306.631	0.469
sys_bp	125.134	122.00	387.482	1.034
dia_bp	71.198	72.00	185.042	-0.898
hba1c	5.811	5.50	1.148	3.239

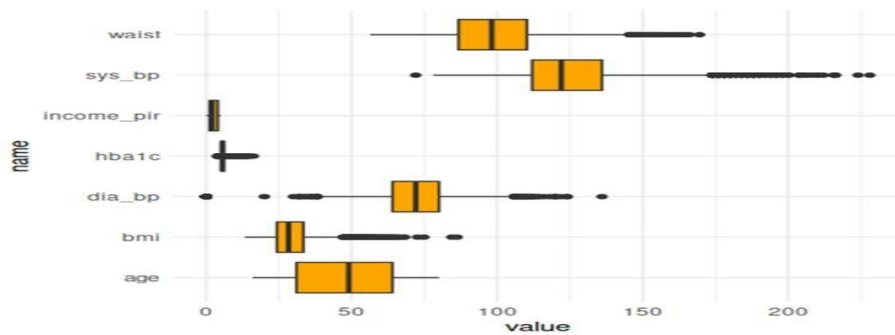
Above table summarizes the selected numeric variables. The sample has an average age of 48.13 years and an average BMI of 29.46, suggesting an overweight population. HbA1c shows strong positive skewness, indicating that a small number of individuals have very high glucose levels. Overall, the variables show meaningful variation, supporting their use in clustering analysis.

2.2 Histograms



Above histograms show the distribution of key variables. BMI, HbA1c, and systolic blood pressure are right-skewed, indicating a small number of individuals with high values. Diastolic blood pressure and waist are more normally distributed, while age is fairly spread across the sample. Overall, the data shows variation and skewness, supporting further analysis for clustering.

2.3 Boxplots for Outlier Detection



Above, boxplots show the distribution and variability of each variable. BMI, systolic blood pressure, and waist exhibit wider spreads and several outliers, indicating variability and the presence of extreme values. HbA1c and income are more concentrated with limited spread, while age shows moderate variability. Overall, the plots highlight differences in dispersion and the presence of outliers across variables.

2.4 Check Missing Data (NAs)



Above figure shows the percentage of missing values across features. Income-to-poverty ratio and blood pressure variables have the highest missingness, while BMI, HbA1c, and waist have moderate levels. Age and sex have little to no missing data. This indicates that missing values are concentrated in a few key variables and should be handled before further analysis.

2.5 Median Imputation for Missing Numeric Values

Median imputation was applied to handle missing values because several variables in the dataset (e.g., BMI, blood pressure, and HbA1c) exhibit skewed distributions and contain outliers.

Unlike the mean, the median is robust to extreme values, meaning it is not heavily influenced by unusually high or low observations. For example, HbA1c shows strong right skewness, indicating a small number of individuals with very high values. Using the mean in such cases would shift the imputed values upward and distort the distribution.

By using the median, the imputed values better represent the typical observation in the data without being affected by outliers. This helps preserve the overall distribution and prevents bias in subsequent analyses such as clustering.

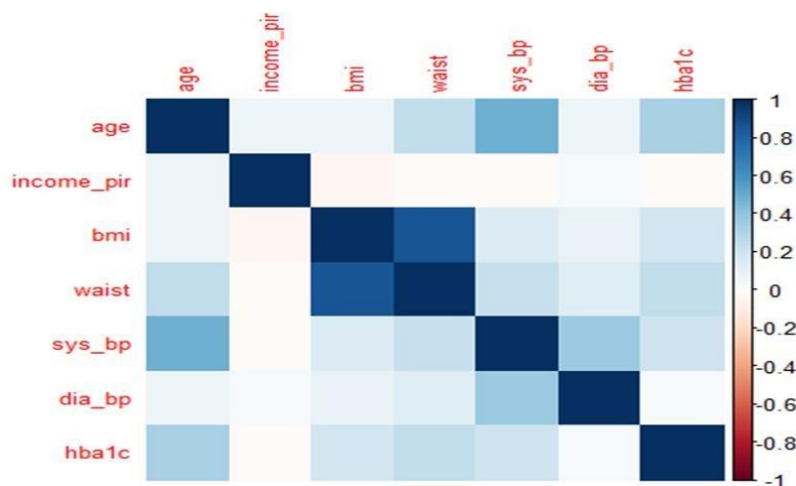
Median Imputation for Missing Numeric Values

Summary Statistics After Median Imputation

Variable	Mean	Median	Min	Max
age	48.13	49.00	16.0	80.0
income_pir	2.45	2.07	0.0	5.0
bmi	29.44	28.30	13.2	86.2
waist	99.26	98.10	56.4	169.5
sys_bp	124.78	122.00	72.0	228.0
dia_bp	71.29	72.00	0.0	136.0
hba1c	5.80	5.50	3.8	16.2

After median imputation, the summary statistics remain largely consistent with the original data. Central tendencies are stable, indicating that the imputation did not significantly distort the distributions. This suggests that median imputation effectively handled missing values while preserving the overall structure of the dataset.

2.6 Correlation Matrix



2.7 Highly Correlated Variables

```
print(high_corr)
```

```

row col
waist 4 3
bmi   3 4
```

2.8 Data Scaling

Check variances

age	income_pir	bmi	sys_bp	dia_bp	hba1c
383.592104	2.267213	54.808526	344.645463	164.178919	1.095336

Check for any NA after scaling

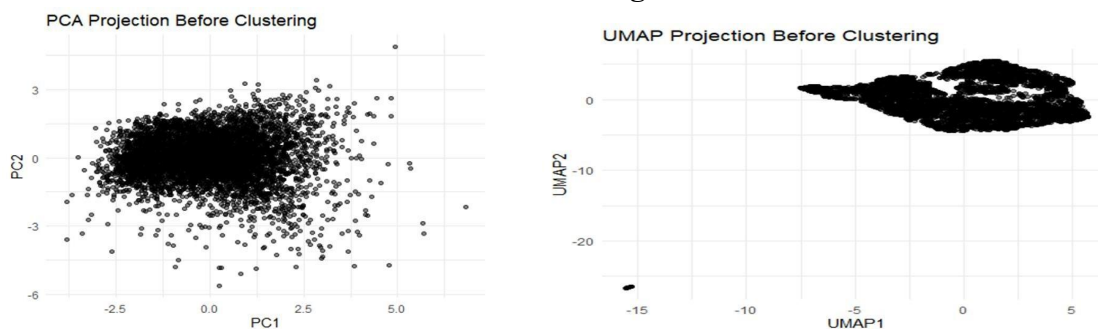
age	income_pir	bmi	sys_bp	dia_bp	hba1c
0	0	0	0	0	0

All numeric variables were standardized using z-score normalization so that each feature has a mean of zero and a standard deviation of one. This was necessary because the variables are measured on different scales, and clustering methods rely on distance calculations. Scaling ensures that no variable dominates the analysis. Categorical variables such as sex were excluded, and highly correlated variables (e.g., waist and BMI) were removed to avoid redundancy.

2.9 Dimensionality Reduction and Visual Exploration

Dimensionality reduction was used to visually examine whether the dataset showed any natural cluster structure before applying clustering algorithms. Principal Component Analysis (PCA) was first applied to the standardized dataset to project the variables into two dimensions while preserving as much overall variation as possible. In addition, UMAP was used to provide a nonlinear two-dimensional projection that can better capture local patterns among observations.

PCA and UMAP visualization before clustering



The PCA plot shows one dense, overlapping cloud with a few outliers, suggesting no clear separation before clustering. The UMAP plot shows more nonlinear structure and a few isolated points, but the groups are still not clearly separated.

3. Methodology

This project used clustering methods to identify natural groups in the NHANES health dataset. The analysis used continuous variables such as age, income, BMI, blood pressure, and HbA1c. Before clustering, missing values were handled, and the selected variables were scaled so that all variables contributed fairly to the distance and similarity calculations. Scaling was important because variables such as income, blood pressure, and HbA1c are measured on different scales.

Three clustering methods were used: DBSCAN, Gaussian Mixture Model, and K-means. These methods were selected because they use different approaches to grouping data. DBSCAN is density-based, GMM is probability-based, and K-means is distance-based. Comparing these methods helped us understand whether the dataset had clear distance-based clusters, dense groups, overlapping groups, or possible outliers.

3.1. DBSCAN (Density Based Spatial Clustering of Applications with Noise)

DBSCAN was selected because it can find clusters based on dense areas of data and can identify noise or outliers. It works by grouping points that are close together while labeling points in sparse areas as noise. The main parameters are *eps*, which controls the neighborhood distance around each point, and *minPts*, which controls the minimum number of points needed to form a dense region. Different *eps* and *minPts* values were tested to find a reasonable clustering solution. DBSCAN is useful because it does not require the number of clusters in advance and can detect unusual observations. However, it is sensitive to parameter choices and may not work well when clusters have very different densities.

3.2 Gaussian Mixture Models (GMM)

GMM was selected because it is useful when clusters may overlap. It works by assuming that the data come from a mixture of normal distributions, where each distribution represents a cluster. Instead of assigning each observation strictly to one cluster at first, GMM estimates the probability that each observation belongs to each cluster. The main parameter is the number of components, or clusters. Different models were compared using BIC, AIC, and clustering validation metrics. GMM is flexible and provides probability-based cluster membership, but it depends on distributional assumptions and can be sensitive to starting values and model structure.

3.3. K-Means Clustering

K-means was selected because it is simple, commonly used, easy to interpret, and suitable for continuous numeric variables. It works by assigning each observation to the nearest cluster center, called a centroid. After the observations are assigned, the centroid is recalculated, and this process continues until the cluster assignments become stable. The main parameter is *K*, or the number of clusters. Several *K* values were tested using the Elbow Method, Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. K-means is fast and useful for creating clear cluster profiles, but it requires choosing the number of clusters in advance and can be affected by outliers and initial centroid placement.

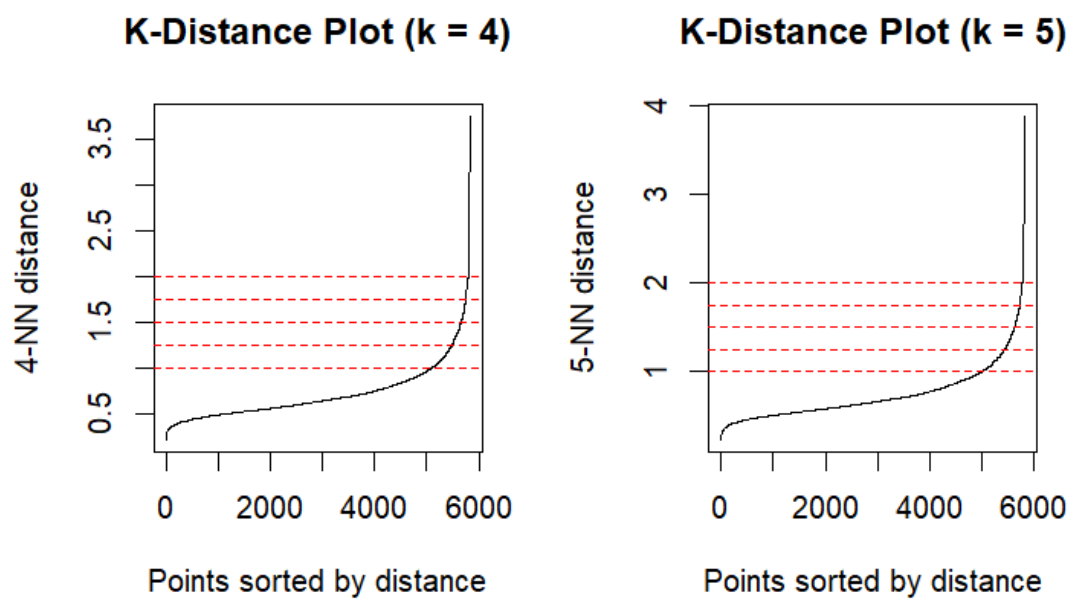
After applying all three methods, the results were compared using validation metrics and practical interpretation. The Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index were used to evaluate cluster quality. In addition, the cluster sizes and health profiles were examined to determine whether the results were meaningful. The final method was selected based on both statistical performance and how useful the clusters were for understanding health profiles in the dataset.

4. Results

4.1 DBSCAN clustering

K Distance plot is used to guide the epsilon value. The plot sorts all observations by their distance to their k-th nearest neighbor. At the elbow, or point where the curve bends, the natural density threshold and starting point for the epsilon value is indicated. To check sensitivity of the elbow to the choice of k, the plot is generated at two values of k.

K Distance plot to determine the best epsilon at elbow



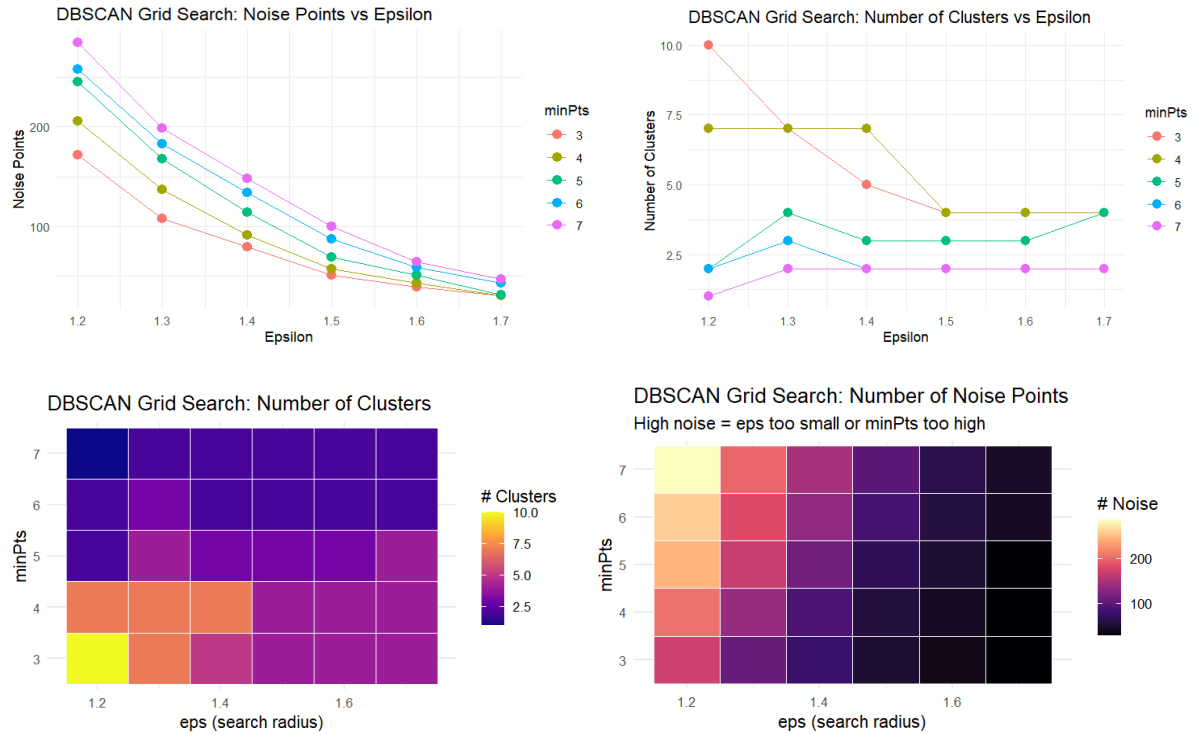
Both plots show an elbow at an approximate value of 1.5, which is consistent across the two k values tested. This consistency indicates the elbow is not an artifact of the choice of k and suggests the natural density threshold lies at around 1.5.

Tests range of epsilon and minPts values around elbow to determine optimal sensitivity

	eps	minPts	n_clusters	n_noise		eps	minPts	n_clusters	n_noise		eps	minPts	n_clusters	n_noise
1	1.2	3	10	172	11	1.6	4	4	43	21	1.4	6	2	134
2	1.3	3	7	108	12	1.7	4	4	30	22	1.5	6	2	87
3	1.4	3	5	79	13	1.2	5	2	245	23	1.6	6	2	59
4	1.5	3	4	51	14	1.3	5	4	168	24	1.7	6	2	43
5	1.6	3	4	39	15	1.4	5	3	114	25	1.2	7	1	285
6	1.7	3	4	30	16	1.5	5	3	69	26	1.3	7	2	199
7	1.2	4	7	206	17	1.6	5	3	51	27	1.4	7	2	148

8	1.3	4	7	137	18	1.7	5	4	31	28	1.5	7	2	100
9	1.4	4	7	91	19	1.2	6	2	258	29	1.6	7	2	64
10	1.5	4	4	57	20	1.3	6	3	183	30	1.7	7	2	47

Visuals of DBSCAN grid search results



A grid search was conducted across a range of epsilon values from 1.2 to 1.7 and minPts values of 3 to 7. Although it subverts the standard rule of minPts being the number of dimensions plus one, a minPts of 5 was chosen. This is because the expected minPts = 7 consistently produced only one or two clusters when tested. At eps 1.5 and minPts 5, DBSCAN produced a stable three cluster solution with 69 noise points (1.18%). An eps of 1.6 was rejected because of the apparent instability of its neighbor 1.7 that produced 4 clusters. In contrast, 1.5 occupies a stable position of three clusters on both sides.

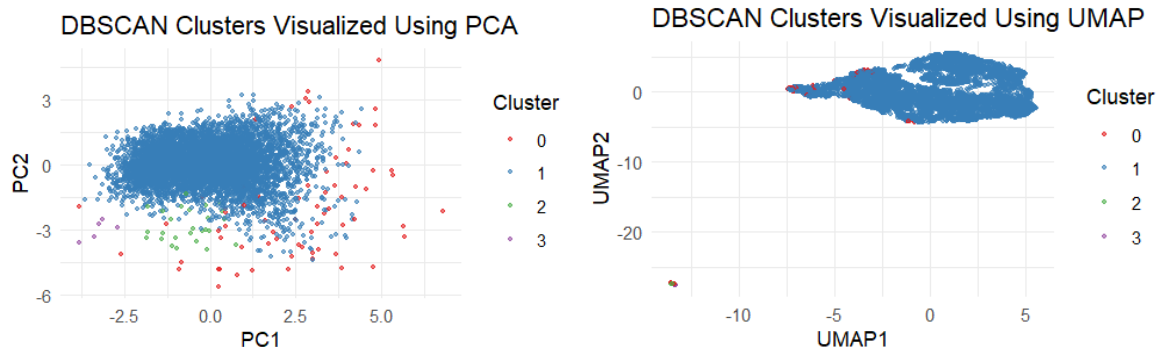
Fit of DBSCAN Model

DBSCAN Model Summary

Objects:	5828	Cluster sizes	
eps =	1.5	Noise (0)	69
minPts =	5	Cluster 1	5728
Distance:	Euclidean	Cluster 2	26
Clusters:	3	Cluster 3	5
Noise points:	69		
Noise %:	1.18%		

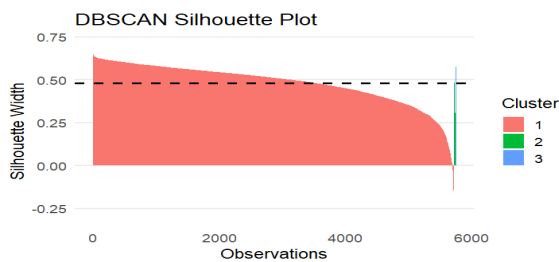
DBSCAN identified three clusters and one noise group. Cluster 1 contains 5,728 observations, over 95% of the data set. Clusters 2 and 3 are significantly smaller at 26 and 5 observations. This leaves 69 observations as noise in Cluster 0. This is not erroneous data but extreme clinical cases that fall outside the minimum density threshold of the other clusters.

DBSCAN Visuals



Both visuals confirm the dominant cluster output. Cluster 1 contains the central mass of the data while the smaller clusters and noise observations surround it on the periphery.

Evaluation metrics of DBSCAN



Per-Cluster Silhouette Distribution

Cluster	N	Mean	Median	Min	Max	% Negative
Cluster 1	5728	0.4815	0.5109	-0.3074	0.6514	0.51
Cluster 2	26	0.3624	0.4103	0.0419	0.5259	0.00
Cluster 3	5	0.6315	0.6258	0.5743	0.7097	0.00

A score of 0.4811 reflects moderate separation with the observations ostensibly matched to their cluster but the boundaries between them are not sharply defined. The per cluster distribution is also uneven. Cluster 1 has a mean of 0.4815 and median of 0.5109 which is essentially the same as the overall average since it contains over 98% of the observations. Cluster 2 shows a lower mean of 0.3624 with no negative widths. This suggests a loose association. And Cluster 3 reports the highest mean at 0.6315, but this is caused by its very small size and is not evidence of strong separation.

Davies-Bouldin Index	Calinski-Harabasz Index	Silhouette Score
0.7432	96.6371	0.4811

The three validation metrics give a consistent account of the DBSCAN model. The silhouette score of 0.4811 portrays a moderate cohesion within the clusters but indicates that the boundaries between them are not readily defined. The Davies-Bouldin Index, at 0.7432, falls in the moderate range. It is not performing badly according to this metric, the value is being greatly steered by the construction of the two smaller clusters (26 and 5), which are put together very densely, shaping the score without reflecting any meaningful cluster architecture across the dataset as a whole. The Calinski-Harabasz Index score, 96.6371, is the revealing of the three. Since this index scales the ratio of between cluster to within cluster dispersion and is also sensitive to cluster balance, the value it awarded DBSCAN confirms that it did not portion the observations well.

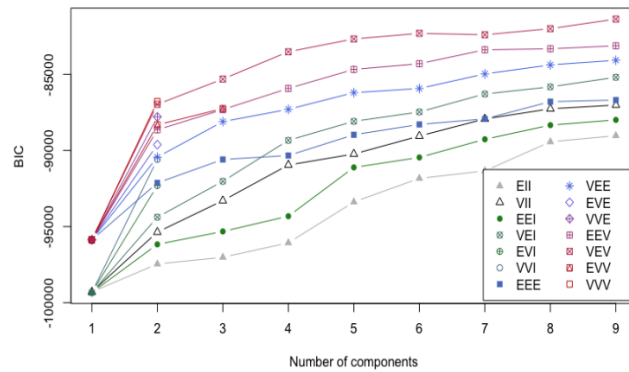
DBSCAN was evaluated across a systemic grid search of 30 parameter combinations and cross checked against alternative candidate fits. Across all configurations, the algorithm either produced a single dominant cluster containing more than 95% of the observations with only trivial satellite groups or collapsed into two clusters with similar imbalances. At no time did DBSCAN produce a substantive multi-cluster structure with adequately sized groups of clusters. This seems to indicate that the NHANES dataset does not contain well separated groups that DBSCAN is able to detect. This also suggests that the health profiles are on a spectrum rather than discrete density-based populations. Given the continuity, the model approaches by Gaussian Mixture Models and K-Means would be more appropriate for identifying the structures that lie within the data.

4.2 GMM CLUSTERING ANALYSIS

Gaussian Mixture Models (GMM) were implemented as a clustering method in this project, as the NHANES data represents a continuous spectrum of health rather than groups which are clearly separable. The probabilistic cluster membership of GMM lends itself to the overlapping health profiles of the dataset and can model clusters with differing shapes and covariance structures, useful for the moderate skewness and correlated variables seen in the EDA phase. GMM models cluster as an overlap of Gaussian distributions, where each cluster has its own center and spread. Points are assigned to clusters based upon probabilities calculated through the Expectation-Maximization algorithm.

The first step after fitting the GMM model is parameter selection. Specifically, the covariance structures must be compared, which can be done in the *mclust* package. The VEV model had the highest BIC values for any covariance structure from G=3 to G=9, with the VVV model having the best performance when G=2.

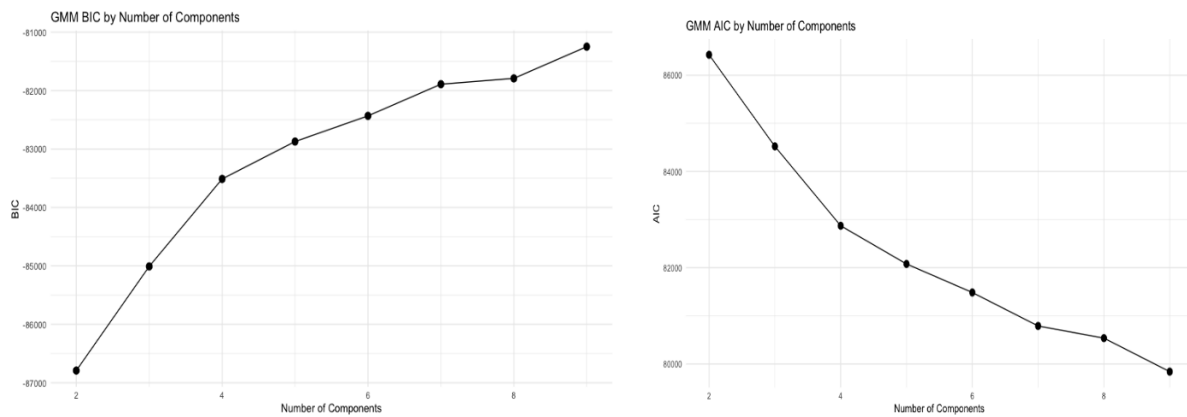
BIC Plot for all GMM Covariance Structures



To determine the optimal number of clusters (G) to be used, several metrics must be considered. The BIC and AIC scores should be considered alongside the Davies-Bouldin and Calinski-Harabasz indices and the average silhouette widths. The table below shows the values of these metrics for two through nine clusters. Individual plots of these metrics follow.

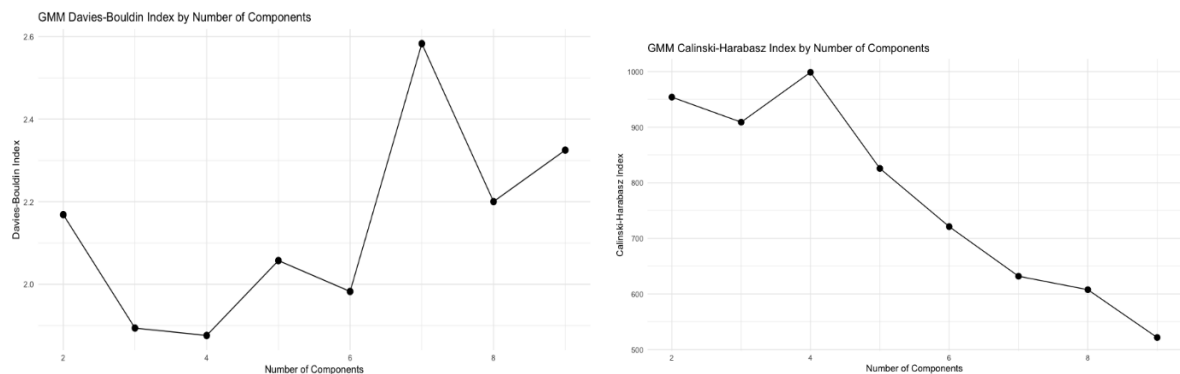
G	Model	BIC	AIC	Silhouette	Davies-Bouldin	Calinski-Harabasz
2	VVV	-86790.89	86424.01	0.2461	2.1686	953.9958
3	VEV	-85008.67	84521.72	0.1667	1.8939	908.9528
4	VEV	-83512.20	82871.84	0.1813	1.8760	998.6605
5	VEV	-82872.22	82078.44	0.1234	2.0574	825.7462
6	VEV	-82433.01	81485.81	0.0997	1.9826	721.0337
7	VEV	-81891.03	80790.41	0.0856	2.5829	631.8080
8	VEV	-81791.56	80537.52	0.0900	2.2003	607.6264
9	VEV	-81248.89	79841.42	0.0476	2.3250	521.5428

GMM BIC and AIC Values



The first significant elbow in this BIC plot occurs at $G=4$. Similar to the BIC plot, this AIC plot also shows an elbow at $G=4$, showing that the AIC and BIC metrics are in good agreement.

Davies-Boldin Index and Calinski-Herbasz Index



In the above figure, the Davies-Bouldin index will be minimized at the optimal number of clusters. This plot shows us a global minimum at $G=4$. The Calinski-Harabasz metric will be a maximum at the optimal G . This plot shows a maximum value at $G=4$.

While the BIC and AIC curves show continued improvement up to $G=9$, the first significant change in slope (the 'elbow') occurs for both plots at $G=4$. This is corroborated by the Calinski-Harabasz index, which reaches its global maximum at $G=4$, and the Davies-Bouldin index, which is lowest at $G=4$. Based on these results, $G=4$ was selected, with the VEV covariance structure, which was shown to have the highest BIC at $G=4$.

Metrics for GMM model with 4 clusters (VEV covariance structure)

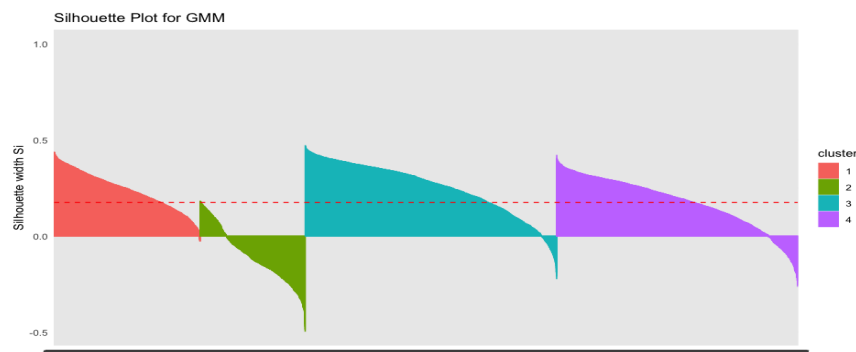
After identifying the appropriate number of clusters, the cluster quality metrics were recorded for $G=4$, seen in the table below:

BIC	AIC	Silhouette	Davies-Bouldin	Calinski-Harabasz
-83512.20	82871.84	0.1746	1.9078	991.787

Below the silhouette scores can be seen for each cluster, along with the size of each cluster:

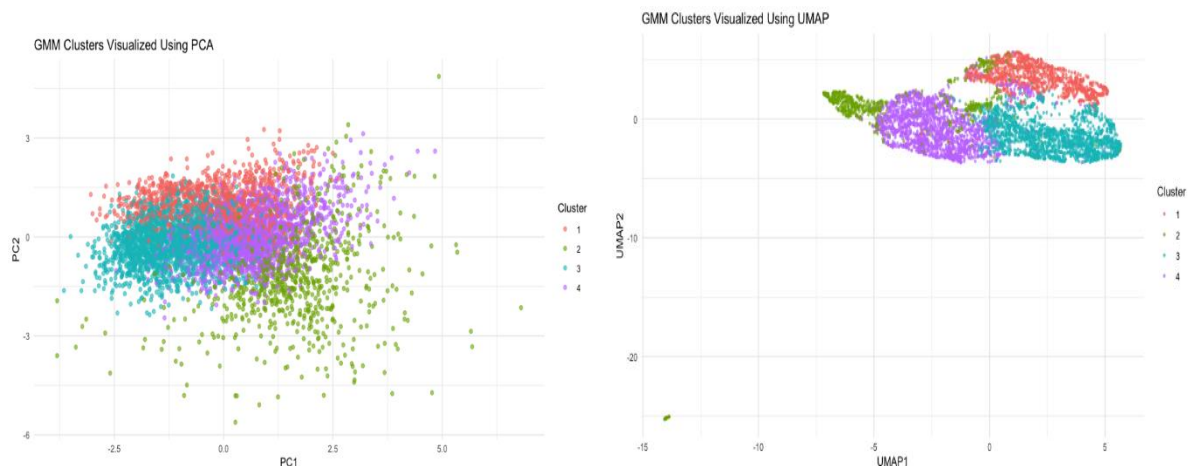
Cluster Number	Number of Observations	Average Silhouette Width
1	1145	0.2395428
2	823	-0.1011910
3	1970	0.2515514
4	1890	0.1752064

GMM Silhouette Plot



The silhouette plot shows us that cluster two has a negative average silhouette score, meaning that many observations in this cluster were closer to observations in neighboring clusters than to their own assigned cluster. This suggests that cluster two may not represent a distinct health profile.

GMM PCA and UMAP Visualization



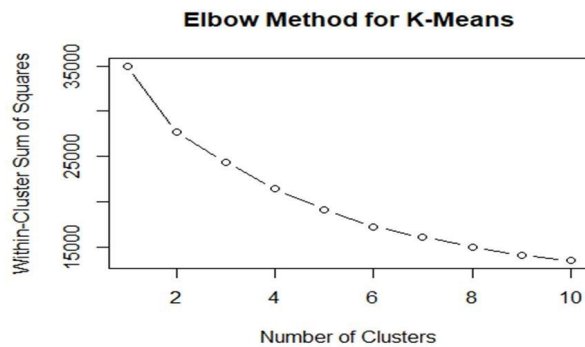
In the above plot, The GMM plot shows four clusters with noticeable overlaps in the PCA space. Cluster 2 is more spread out, while the other clusters are more concentrated near the

center. Likewise, the UMAP visualization shows us that cluster two appears split into two subgroups, indicating that the GMM model may have forced observations together which may naturally have belonged to other groups. This also reflects the negative average silhouette score for cluster two.

4.3 K-MEANS CLUSTERING ANALYSIS

K-means clustering was applied to the standardized continuous variables: age, income, BMI, systolic blood pressure, diastolic blood pressure, and HbA1c. The method groups observations by minimizing within-cluster variation and assigning cases to the nearest centroid using Euclidean distance.

Elbow Method for Choosing K

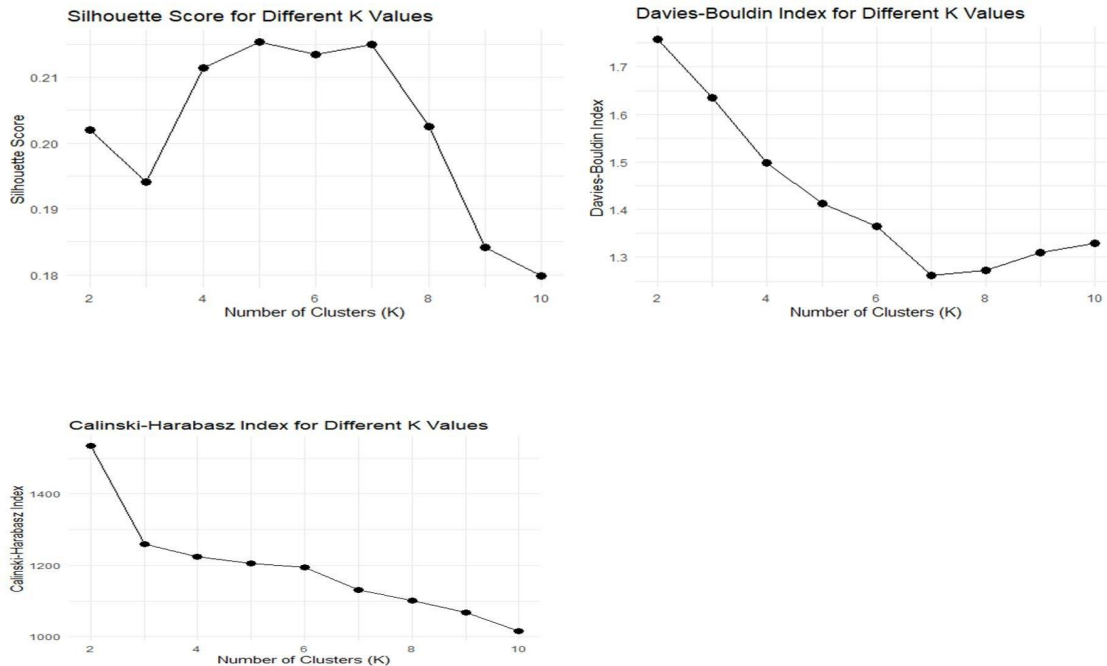


While the DBSCAN k-distance plots showed a reasonably consistent elbow around 1.5 for selecting epsilon, the K-means elbow plot did not show a sharp or clearly defined elbow for selecting K. Instead, the K-means curve showed a gradual decline in within-cluster sum of squares, suggesting that the data may contain overlapping cluster structures rather than clearly separated groups. Therefore, formal validation metrics were used to select K more rigorously.

Validation Metrics for Different K Values

	K	Silhouette_Score	Davies_Bouldin_Index	Calinski_Harabasz_Index
1	2	0.2019203	1.757493	1534.722
2	3	0.1940932	1.635532	1259.597
3	4	0.2113755	1.497466	1223.244
4	5	0.2153430	1.412943	1205.100
5	6	0.2134919	1.364325	1195.268
6	7	0.2149356	1.261704	1131.680
7	8	0.2024621	1.273045	1101.250
8	9	0.1842117	1.308878	1067.824
9	10	0.1798195	1.328546	1015.720

Plot Validation Metrics



The Silhouette Score plot shows that the highest value occurs at $K = 5$. Since a higher Silhouette Score is better, this suggests that the five-cluster solution gives the best balance of within-cluster similarity and between-cluster separation among the tested values. However, the score is still relatively modest, meaning the clusters are not very strongly separated.

The Davies-Bouldin Index plot generally decreases as K increases and reaches its lowest value around $K = 7$. Since a lower Davies-Bouldin Index is better, this metric slightly favors $K = 7$. However, $K = 5$ also performs reasonably well and avoids creating too many clusters.

The Calinski-Harabasz Index plot is highest at $K = 2$ and then declines as K increases. Since a higher value is better, this metric favors a simpler two-cluster solution. However, $K = 2$ may be too broad and may not provide enough meaningful subgroup interpretation.

Based on these mixed validation results, $K = 5$ was selected as a balanced and interpretable solution. Although the Davies-Bouldin Index slightly favored $K = 7$ and the Calinski-Harabasz Index favored $K = 2$, the Silhouette Score was highest at $K = 5$, indicating the best overall balance of cohesion and separation among the tested values. In addition, $K = 5$ provided more meaningful subgroup detail than $K = 2$ while avoiding the added complexity of $K = 7$.

Final K-means Evaluation Metrics

Method	Silhouette	Davies_Bouldin	Calinski_Harabasz
K-means	0.2153	1.4129	1205.1

PCA and UMAP Visualization of K-Means Clusters



The PCA visualization shows that the five K-means clusters are partly separated but still overlap substantially, especially near the center of the plot. This supports the earlier validation results, where the modest Silhouette Score suggested that the clusters were not strongly separated.

The UMAP visualization shows a clearer curved structure, with some clusters forming more visible regions. However, there is still overlap among the groups, indicating that the data contains gradual health and demographic patterns rather than completely distinct clusters

Comparison of all three methods

Statistical comparison

Method	Silhouette Score	Davies-Bouldin Index	Calinski-Harabasz Index	BIC and AIC (GMM only)	Cluster Structure
K-means	0.2153	1.4129	1205.100	N/A	Five reasonably sized clusters
DBSCAN	0.4811	0.7432	96.64	N/A	One large cluster, two tiny clusters, and noise
GMM	0.1746	1.9078	991.79	-83512.20 (BIC) -82871.84 (AIC)	Four clusters, but weaker separation

Practical interpretation

Method	Main Strength	Main Limitation	Final Use
K-means	Produced balanced and interpretable health profiles	Requires choosing the number of clusters in advance	Selected as the final method
DBSCAN	Detected dense regions and outliers	Created one very large cluster and very small remaining clusters	Used for comparison and outlier detection
GMM	Allowed overlapping probability-based clusters	Showed weaker separation and one poorly defined cluster	Used for comparison

Overall, the three clustering methods produced different results. DBSCAN had the strongest statistical scores, with the highest Silhouette Score and lowest Davies-Bouldin Index, but its cluster structure was less balanced because it produced one large cluster, two tiny clusters, and noise points. GMM allowed probability-based overlapping clusters, but it showed weaker separation and less clear cluster profiles. K-means had more modest validation scores, but it produced five reasonably sized and interpretable health profiles. Therefore, K-means was selected as the final method because it offered the best balance between statistical performance and practical interpretability.

5. Cluster Profiling and Discussion

K-means was selected as the best performing method because it produced five reasonably sized and interpretable clusters. The NHANES variables were continuous and represented gradual differences in health risk rather than sharply separated groups, which made K-means useful for creating practical health profiles. DBSCAN had stronger validation metrics, but its results were highly imbalanced, with most observations assigned to one large cluster and only a few observations assigned to smaller clusters. GMM produced more balanced clusters than DBSCAN, but the cluster separation was weaker, and one cluster showed poor silhouette performance. Since the goal of the project was to identify meaningful health profiles rather than only maximizing validation scores, K-means was the most appropriate final method.

Cluster Size and Percentage

K-means Cluster Size and Percentage

kmeans_cluster	n	percent
1	1317	22.60
2	314	5.39

3	1709	29.32
4	1598	27.42
5	890	15.27

The K-means model divided the dataset into five clusters, with Clusters 3 and 4 being the largest groups. Cluster 2 is the smallest group.

Summary Statistics and Cluster Profiles

K-means Cluster Profile Summary

kmeans_cluster	n	mean_age	mean_income_pir	mean_bmi	mean_systolic_bp	mean_diastolic_bp	mean_hb_a1c
1	1317	49.38	4.62	27.81	120.37	72.13	5.59
2	314	60.29	2.17	32.51	129.50	69.73	9.23
3	1709	30.48	1.66	25.67	111.58	65.46	5.31
4	1598	66.01	1.89	27.80	141.16	75.24	5.85
5	890	43.77	1.90	40.97	125.59	74.69	5.74

This table summarizes the five K-means health profiles. Cluster 1 is higher income with moderate health risk, Cluster 2 shows the highest diabetes risk, Cluster 3 appears youngest and lowest risk, Cluster 4 reflects older adults with higher blood pressure, and Cluster 5 represents the highest BMI or obesity-related risk group.

Sex Distribution by Cluster

kmeans_cluster	sex	n	percent
1	Female	645	48.97
1	Male	672	51.03
2	Female	141	44.90
2	Male	173	55.10
3	Female	922	53.95
3	Male	787	46.05
4	Female	779	48.75
4	Male	819	51.25

5	Female	522	58.65
5	Male	368	41.35

This table shows the sex distribution within each K-means cluster. Clusters 1, 2, and 4 have slightly more males than females, while Clusters 3 and 5 have more females than males. Cluster 5 has the largest female share at 58.65%, while Cluster 2 has the largest male share at 55.10%. Overall, the clusters are not extremely imbalanced by sex, but there are some noticeable differences across groups.

Sex Distribution Plot



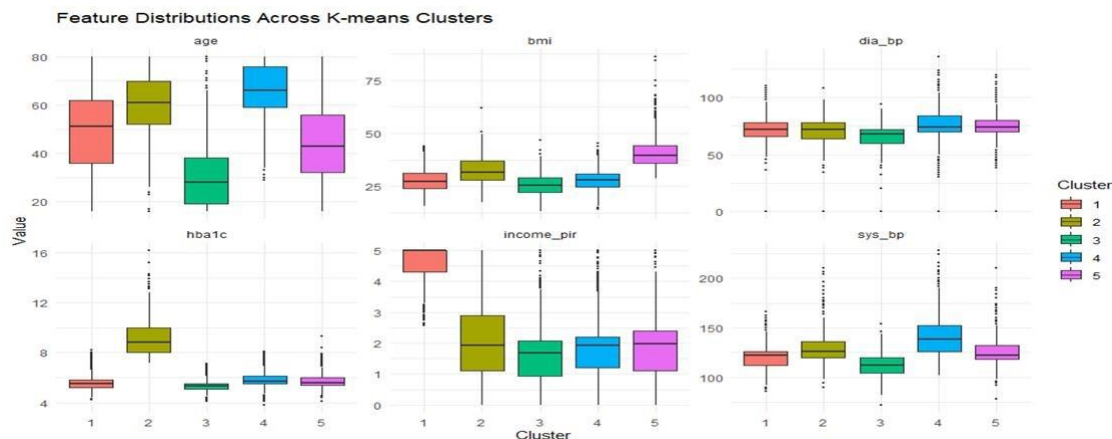
This plot shows the percentage of males and females in each K-means cluster. Clusters 1, 2, and 4 have slightly higher male percentages, while Clusters 3 and 5 have higher female percentages. Cluster 5 shows the clearest difference, with females making up a larger share of the cluster. Overall, sex distribution varies across clusters, but most clusters are still relatively balanced.

Feature Comparison Bar Plot



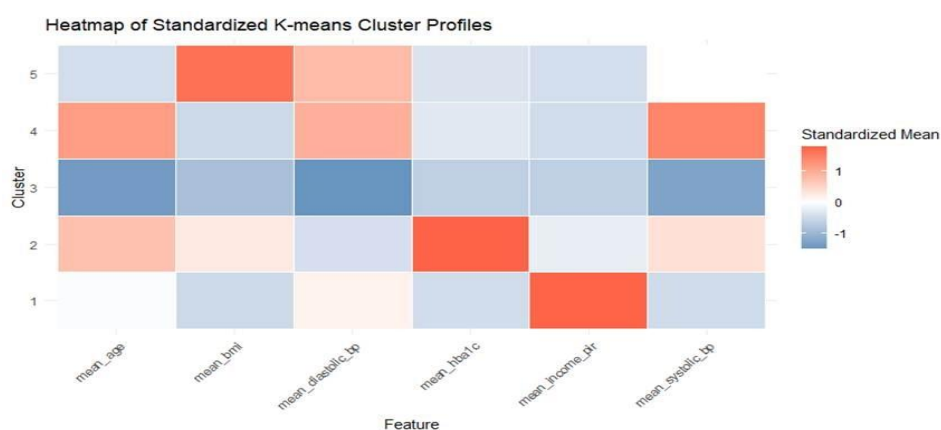
This plot compares the average values of key variables across the five K-means clusters. Cluster 2 appears to have the highest average HbA1c and relatively high BMI, suggesting a higher diabetes or metabolic risk group. Cluster 4 has the highest average age and systolic blood pressure, suggesting an older group with greater cardiovascular risk. Cluster 5 has the highest BMI, indicating an obesity-related risk group. Cluster 1 has the highest income-to-poverty ratio, while Cluster 3 appears younger and has lower BMI and blood pressure, suggesting a relatively lower-risk group.

Feature Distribution Boxplots



This boxplot shows how each health variable is distributed across the five K-means clusters. Cluster 4 has the highest age and systolic blood pressure, suggesting an older group with higher cardiovascular risk. Cluster 2 has the highest HbA1c levels, indicating a higher diabetes-related risk group. Cluster 5 has the highest BMI, suggesting an obesity-related risk group. Cluster 1 has the highest income-to-poverty ratio, while Cluster 3 appears younger and generally lower on BMI, blood pressure, and HbA1c, suggesting a relatively lower-risk group.

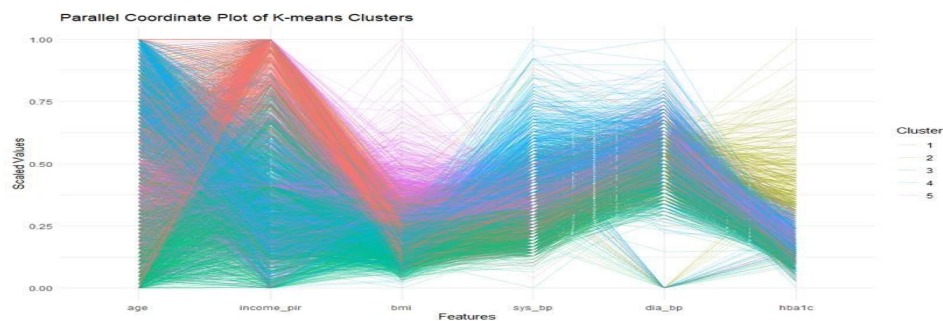
Heatmap of Standardized Cluster Profiles



This heatmap shows which features are relatively high or low in each K-means cluster after standardization. Red means the cluster is above the overall average for that feature, while blue means it is below average. Cluster 2 stands out with the highest HbA1c, suggesting higher diabetes-related risk. Cluster 4 shows higher systolic blood pressure and older age, suggesting greater cardiovascular risk. Cluster 5 has higher BMI, while Cluster 1 has the highest income-

to-poverty ratio. Cluster 3 is mostly blue, meaning it is generally lower across several health-risk indicators.

Parallel Coordinate Plot



This parallel coordinate plot shows how observations in each K-means cluster differ across the scaled variables. Cluster 1 is clearly higher on income, while Cluster 2 appears higher on HbA1c, suggesting greater diabetes-related risk. Cluster 4 tends to be higher on age and blood pressure, suggesting an older cardiovascular-risk group. Cluster 5 is higher on BMI, indicating an obesity-related profile. Overall, the plot confirms that the clusters differ across several health and demographic features, although there is still some overlap among groups.

Business/Scientific Meaning of the K-means Clusters

Cluster 1 appears to represent a higher-income, moderate-health group. This cluster has the highest income-to-poverty ratio and moderate values for BMI, blood pressure, and HbA1c. Scientifically, this group may represent individuals with better socioeconomic status and relatively manageable health risk. From a public health perspective, this group may have better access to healthcare, nutrition, and preventive services.

Cluster 2 appears to represent a high diabetes-risk group. This cluster has the highest HbA1c level and relatively high BMI. Since HbA1c is an important marker of blood sugar control, this cluster may represent individuals with higher risk of diabetes or poor glucose regulation. This group would be important for diabetes screening, lifestyle intervention, and chronic disease management.

Cluster 3 appears to represent a younger and lower-risk group. This cluster has the lowest average age and generally lower BMI, blood pressure, and HbA1c values. Scientifically, this cluster may represent a healthier or lower cardiometabolic-risk population. This group may need general preventive care rather than intensive medical intervention.

Cluster 4 appears to represent an older cardiovascular-risk group. This cluster has the highest average age and the highest systolic blood pressure. Higher systolic blood pressure is often associated with increased cardiovascular risk, especially among older adults. This group may benefit from blood pressure monitoring, hypertension prevention, and cardiovascular health programs.

Cluster 5 appears to represent an obesity-related risk group. This cluster has the highest BMI, suggesting that body weight is the main feature distinguishing this group. Even if HbA1c

and blood pressure are not the highest, high BMI can increase future risk for diabetes, hypertension, and other chronic conditions. This group may benefit from weight management, nutrition support, and physical activity interventions.

Overall, these clusters show that health risk in NHANES data is not the same for everyone. Some groups are mainly separated by income, some by diabetes risk, some by age and blood pressure, and some by BMI. The scientific value of clustering is that it helps identify different population health profiles that may need different types of prevention or intervention.

6. Conclusion

This project used K-means, DBSCAN, and Gaussian Mixture Models to identify health-related clusters in the NHANES dataset. Before clustering, the data were cleaned by handling missing values, removing redundant variables, and scaling the numeric features so that each variable contributed fairly to the analysis.

Among the three methods, K-means was selected as the final clustering method because it produced the most balanced and interpretable cluster solution. The five K-means clusters showed meaningful differences in age, income-to-poverty ratio, BMI, blood pressure, and HbA1c. Sex was not used to form the clusters, but it was included in the profiling stage to better describe the demographic composition of each group. The sex distribution showed that most clusters were relatively balanced, although some clusters had slightly higher male or female representation.

Overall, the results suggest that the sample includes several distinct health-risk profiles, including lower-risk individuals, groups with higher diabetes-related risk, groups with higher cardiovascular risk, and groups with obesity-related risk. These cluster profiles may help researchers and public health professionals identify groups that could benefit from more targeted prevention, screening, or intervention programs. Future research could include additional health, lifestyle, demographic, and behavioral variables to improve cluster interpretation and examine whether these clusters are associated with future health outcomes.